

Md. Hafijur Rahman

MSc student in Biomedical Technology with integrated expertise in experimental and computational biology — virology, vaccine development, molecular biology, molecular modelling and multi-omics.

Education

Ongoing	M.Sc. in Biomedical Technology Dokuz Eylül University, Türkiye
2025	M.Sc. in Biotechnology Bangladesh Agricultural University · CGPA 3.73 / 4.00
2022	B.Sc. in Biotechnology & Genetic Engineering Khulna University, Bangladesh · CGPA 3.67 / 4.00

Project works

2023–24	Attenuated fowlpox vaccine from a wild isolate MSc thesis (BAURES). Virus culture & attenuation in embryonated duck eggs, confirmation at each passage, adjuvanted formulation, chicken trial and antibody analysis. Supervisors: Dr. Shahidul Haque, Dr. Md. Bahanur Rahman.
2021–22	Drug discovery against dengue virus targeting NS2B/NS3 (<i>in silico</i>) Literature review, drug screening, structural analogues, <i>in silico</i> and statistical analysis. Supervisor: Khandker Khaldun Islam, Khulna University.
2021	Bioactive compounds blocking the α-dystroglycan receptor to suppress Lassa virus (<i>in silico</i>) Project framework design, docking, ADMET, molecular dynamics simulation and statistical analysis.

B.Sc.

Alzheimer's checkpoints & *Rauvolfia serpentina* phytochemicals against BACE1 (*in silico*)

Undergraduate thesis, Khulna University. Supervisors: Khandker Khaldun Islam, Dr. S M Abdul-Awal.

Publications

2023 ***Phyllanthus emblica* (Amla) methanolic extract regulates checkpoints in 15-lipoxygenase mediated inflammopathies**

doi.org/10.1016/j.jsps.2023.06.014

2023 ***In vitro* and *in silico* investigation of garlic's (*Allium sativum*) bioactivity against 15-lipoxygenase mediated inflammopathies**

doi.org/10.34172/jhp.2023.31

2021 **Phytochemical ligands blocking SARS-CoV-2 S-protein attachment to ACE2: an in-silico approach**

doi.org/10.5455/javar.2021.h481

Presentations

2025 **Integrated CRISPR–microfluidic platform for point-of-care pathogen detection**

Oral · DEUISGR 2025, Dokuz Eylül University

2020 **Molecular docking of phytochemicals targeting HER-2 in breast cancer**

Poster · IPPC 2020 · Outstanding Poster (243 submissions, Life Science)

Skills

Wet lab

Virus propagation & attenuation

Vaccine formulation & trials

Antibody analysis

ELISA

PCR

DNA/RNA extraction

Restriction digestion

Agarose gel electrophoresis

Microbial culture

MIC & disc diffusion

Biochemical assays

Chromatography

Computational

Python

PyTorch

Deep learning

ESM-2

Immunopeptidomics

Molecular docking & MD

Multi-omics

Statistics & visualisation

Docker

Git

Awards

2024 Türkiye Bursları Scholarship

2023 National Science and Technology (NST) Fellowship, Bangladesh

2018–22 Merit Scholarships, Khulna University (2018, 2019, 2021, 2022)

Training & languages

Seminars on biomedical research & career development (2025) and single-cell/spatial transcriptomics (2022); workshop on RNA-seq data analysis (2022).

Bengali – native

English – C1

Turkish – C1